# CV 1000-Puzzle Piece Project: Puzzle Reconstruction by Images

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***Abstract*—This report details the approach taken to segment puzzle pieces from visual images, identify the four corners of each piece, and match the contours and images of the pieces to reconstruct the original puzzle via adjacency graph construction. It focuses on the early successes of the project and the struggles seen in final reconstruction, as well as details the variety of approaches taken to address the difficulty of the task at hand.**

## I. Pipeline Overview

This section provides the high-level design of the system.

- **Design Rationale:** The intention was to utilise state-of-the-art methods to approach the difficult nature of the task. The design was initially based on the most successful U-Net structures found in our initial labs from the course and then to build upon and improve these as time progressed using online material as guidance in improvements.
- **System Architecture:** Initially, the images are fed into a U-Net model with an EfficientNet-B4 encoder, trained to isolate individual puzzle pieces from the background by predicting binary masks. This allowed for lightweight training with adequate accuracy. Second, a specialized U-Net (EfficientNet-B0) regression model is used to predict corner heatmaps. These predictions are refined via Harris Corner detection and a custom sanitization algorithm that snaps corners to the nearest contour pixels to ensure geometric validity. Third, we find piece adjacency by extracting normalized side contours and average colours, scoring pairwise matches with a weighted shape-colour metric, and enforcing a strict one-to-one global locking mechanism to prevent conflicting connections. This part of the pipeline struggled greatly (likely due to challenges in feature extraction) and is where the largest bottleneck of performance was found. Finally, we reconstruct the puzzle by iteratively aligning pieces via estimated affine transformations, using a "max-blend" compositing technique to merge overlapping regions onto a unified canvas

## II. Methods

### A. Segmentation

The segmentation pipeline involved an iterative process, moving from traditional computer vision techniques to deep learning architectures.

- Initial Approach: Initially, we attempted standard image processing techniques, specifically Canny Edge Detection combined with K-Means Clustering and Adaptive Thresholding (as explored in Lab 1). While computationally inexpensive, this approach proved highly volatile. It failed to distinguish puzzle pieces from the background when shadows were present or when the piece colour matched the background table. The irregular tabs also made standard contour algorithms unreliable, resulting in fragmented masks.
- Second Design: U-Net with MobileNetV2 We realised that we need semantic understanding in the model, and we shifted to a Deep Learning approach using a U-Net architecture with a MobileNetV2 encoder.
  This model provided a significant improvement over traditional methods, successfully learning the general shape of puzzle pieces. Although MobileNetV2 is efficient, it is a lightweight model designed for speed rather than maximum depth. We observed that it sometimes struggled with the pixel-level precision required to cleanly capture tabs. The resulting masks were pretty good, but we wanted higher accuracy in details.
- Final Design: U-Net with EfficientNet-B4 To address the precision limitations, the final system upgraded the encoder to EfficientNet-B4. EfficientNet-B4 uses compound scaling which balances depth, width, and resolution. This deeper encoder captured some textures that MobileNet missed and slightly bumped up accuracy. The resulting binary masks showed sharper boundaries and higher Intersection over Union (IoU) scores.

### B. Corner and Contour Extraction

Feature Extraction (Corner Prediction) also considered quite a few approaches, progressing from geometric algorithms to the final deep-learning solution

- Initial Approach: Geometric Contour Approximation The first attempt relied on computational geometry to identify corners from the binary masks generated by the segmentation step. We extracted the contour of the piece using cv2.findContours and attempted to approximate it to a polygon using the Douglas-Peucker algorithm (cv2.approxPolyDP). The algorithm attempts to simplify the jagged contour by reducing the number of vertices. We iterated through epsilon values, shrinking the polygon until it supposedly had only 4 points. This failed on puzzle pieces with tabs and blanks. A standard puzzle piece was complex and this sliced off the tabs or ignored

the holes. Tips of tabs and the corners were mistaken widely.

- Active Contours (Shrinking  Growing) To address the complex shapes, we attempted an Active Contour model.Process: We initialised a bounding box around the piece and shrunk it iteratively. The model minimsed an energy function. This approach famously "got stuck at local minimums The box caused the contour to bridge across the concave holes of the puzzle pieces rather than entering them. It would crop the piece too tightly, cutting off the edges. The growing process would stop before the true corners were found, so we saw that it could not understand that a corner is a specific structural point, not just a high-gradient edge.
- Final Approach: Deep Learning Heatmap Regression Recognising that geometry was insufficient for semantic corner definitions, we treated corner detection as a dense regression problem. We used a U-Net with an EfficientNet-B0 encoder1. Instead of predicting coordinates directly, the network predicts a Heatmap—a $256 \times 256$ probability field where bright "hotspots" indicate corner locations. This started with labelling 200 of the puzzle pieces manually. This proved insufficient so this was increased to 500 pieces out of the total 1000. The raw model output is robust but "fuzzy." To be precise, we implemented 3 post-processing steps. The first was centroid prediction where we used the heatmap peaks to find the approximate corner location. We then used Harris Refinement. This multiplied the heatmap by a cornerHarris response map to prioritize sharp geometric corners over rounded tab tips. Finally we used contour snapping which "snaps" the predicted point to the nearest pixel on the actual mask contour to guarantee the corner lies physically on the piece boundary. The Deep Learning approach performed the best because it learned the semantic appearance of a puzzle corner, making it robust to most shape variations that trapped the geometric algorithms in local minima.

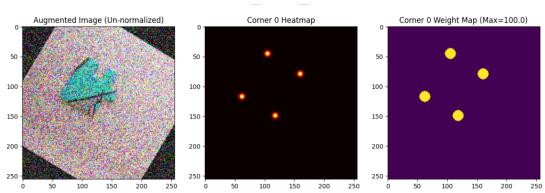


Fig. 1. Sample of the Generated Heatmap for Puzzle Pieces.

### C. Puzzle Graph Construction

The adjacency graph relied on a feature extraction pipeline designed to characterise the pair-wise asymmetry of puzzle tabs.

*1) Feature Extraction:* We implemented a geometric approach that broke each puzzle piece into four discrete sides based on the corner keypoints detected in the previous stage. To ensure strong comparisons, we applied the following normalisation methods:

- **Contour Resampling:** Each side contour was resampled to exactly 50 equidistant points using linear interpolation, creating a fixed-length vector representation for shape comparison.
- **Pose Normalisation:** To achieve translation and rotation invariance, side contours were centered at the origin and rotated such that their endpoints aligned with the X-axis.
- **Side Classification:** We calculated the signed deviation of the contour from the straight line connecting its endpoints. Based on a set threshold (0.03), sides were classified as Flat (Edge), Out (Tab), or In (Hole).
- **Color Features:** To reduce geometrically compatible but visually distinct pieces matching, we extracted the average pixel color of the side edge. This was converted to the CIELAB color space to align with human color perception.

*2) Pair-wise Adjacency and Global Locking:* Adjacency was determined via a weighted scoring function that combined the Euclidean distance between normalised shape vectors and the Delta-E color difference. We enforced logical compatibility constraints, only attempting to match Out tabs to In holes of the opposite orientation.

To combat the "Star Effect"—where multiple pieces erroneously claimed the same neighbor—we implemented a **Global Locking** algorithm. Instead of local greedy selection, we computed all $N^2$ potential match scores, sorted them by quality, and locked pairs in a strict one-to-one manner.

**Critical Evaluation:** Despite these clever feature engineering techniques, the graph construction was an overall failure. The system proved extremely volatile to upstream errors; slight inaccuracies in the corner prediction stage caused tabs to be misclassified as flat edges or distorted the shape descriptors enough to break the matching threshold. This resulted in a sparse graph dominated by isolated clusters rather than a fully connected lattice.

### D. Final Assembly

The assembly stage aimed to physically reconstruct the puzzle by mapping pieces onto a unified 25,000-pixel canvas using the graph.

*1) Mathematical Approach:* We treated the assembly as a sequence of transformations. For every connected pair of pieces, we defined a correspondence between the two corners.

We solved for the optimal Affine Transformation matrix $M \in \mathbb{R}^{2\times 3}$ using the Partial Affine Estimation method (cv2.estimateAffinePartial2D). This method restricted the degrees of freedom to rotation and translation, which is physically accurate for rigid puzzle pieces.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & t_x \\ \sin\theta & \cos\theta & t_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \tag{1}$$

To prevent distortion from false matches, we implemented a check, rejecting any transformation that required scaling the piece by more than 20% larger or smaller.

*2) Stitching and Blending:* The stitching process used a Breadth-First Search (BFS) starting from the most connected anchor piece. To handle the irregular boundaries of the puzzle pieces, we employed a max-blending technique. Instead of overwriting, which often introduced black artifacts from the mask bounding boxes, we used a pixel-wise maximum operation to merge new pieces onto the canvas.

**Evaluation:** While the mathematical framework for stitching was sound, the assembly ultimately failed to produce a coherent global image. The technique relied on a high-fidelity graph. Because the underlying graph was fragmented, the assembly script could only generate small, disjointed islands of 2-3 pieces. Furthermore, the lack of global optimization (e.g., Bundle Adjustment) meant that small errors accumulated over pieces, leading to misalignment in large clusters.

## III. Metrics and Results

### A. Segmentation Performance

The final segmentation model, employing a U-Net architecture with an EfficientNet-B4 encoder, demonstrated exceptional performance on the puzzle dataset.

*1) Quantitative Results:* After 25 epochs of training, the model achieved a **Training IoU of 0.9863** and a **Validation IoU of 0.9787**. The minimal divergence between training and validation scores (0.0076 difference) indicates that the model generalized well without significant overfitting. The final Loss was recorded at 0.0052 (Training) and 0.0081 (Validation).

*2) Qualitative Analysis:* Initial iterations of the pipeline yielded inconsistent results. As shown in Figure 2, early models produced strong masks overall, but often failed to capture the precise boundary of the puzzle tabs or were confused by lighting gradients on the table surface. This resulted in a few jagged corners or missed textures.

To combat this, the final inference stage incorporated Test Time Augmentation **(TTA)**. By averaging predictions across multiple augmented versions (flips and rotations) of the input image, the final model produced consistent binary masks with smoother boundaries. This pixel-level precision was critical, as even small jagged edges would can cause failures in the later stages.

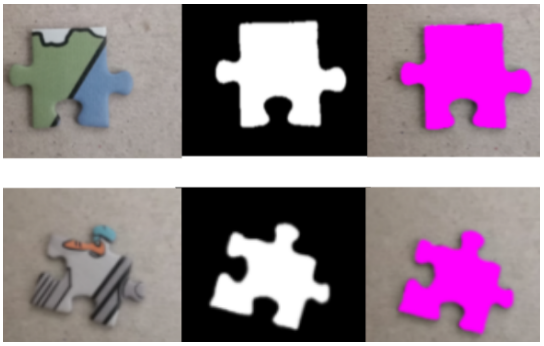

Fig. 2. Examples of inconsistent segmentation from initial model iterations, showing boundary deterioration and noise.

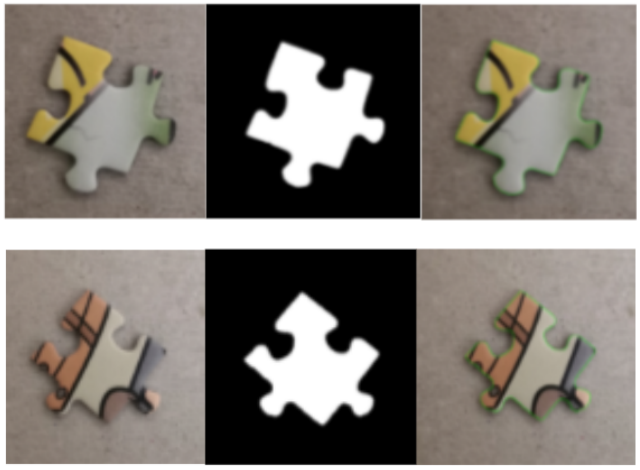

Fig. 3. Final segmentation results (EfficientNet-B4 + TTA). The masks exhibit sharp boundaries and accurate tab isolation.

### B. Corner Detection

*1) Evaluation: Eye Test:* Due to the absence of ground-truth corner coordinates for the unlabelled dataset, formal metrics could not be calculated. Instead, we employed a qualitative evaluation, inspecting a stratified random (yet representative of all piece types) sample of 50 pieces. Success was defined as the detection of exactly four keypoints corresponding to the geometric corners (the armpits of the tabs), while failure was defined as missing corners, false positives on tab tips, or poor localization (major pixels deviation).

*2) Initial Approach: Geometric Contour Evolution:* Our initial attempt utilized an iterative geometric approach inspired by Active Contours (Snakes). The algorithm attempted to fit a 4-sided polygon to the piece boundary by iteratively shrinking a bounding box towards the contour. Initially, there were major successes in bounding the entire piece. However, after shrinking to a box within the borders of the box, major issues occurred.

**Failure:** As shown in Figure 4, this method frequently converged to local minima. The smoothness caused the contour to bridge across the holes of the puzzle piece, while the external energy failed to push the contour into the tab geometries. This resulted in many corners being placed on the tips of tabs rather than the true structural corners, and many bounding boxes getting stuck within the puzzle piece.

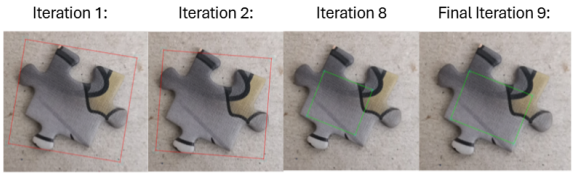


Fig. 4. Failure of the Iterative Geometric approach. The contour (green) gets stuck in a local minimum, cutting off the puzzle tabs (red region) instead of snapping to the true corners.

*3) Final Approach: Deep Heatmap Regression:* The final model (U-Net + EfficientNet-B0) demonstrated significant improvements. By learning the semantic appearance of a corner, it successfully ignored most tab tips.

**Successes:** On standard pieces (maximum 2-tab/), the model achieved near-perfect accuracy, effectively identifying corners even in the presence of noise from segmentation. It was also successful on quite a few of the advanced geometries (Figure 5).

**Failures:** The primary failure mode occurred with outlier shapes, specifically some pieces with 3 or 4 tabs. In these cases, the model often failed to distinguish between the multiple potential corners, sometimes placing keypoints on the tab tips or missing the 4th corner entirely (Figure 6). This suggests the training set was either biased towards standard piece shapes or lacked sufficient data to perfectly capture complicated geometries.

Fig. 5. Final Successful Corner Detections. This highlights some successful detection on all types of pieces.

### C. Graph Connectivity

As ground-truth adjacency data was not provided, we performed a manual audit of the predicted edges.

*1) Precision Analysis:* Visual inspection of the "locked" connections revealed a significant failure in the matching logic. While the system reported high confidence scores (low shape difference), a manual audit of 50 randomly sampled edges showed that many were False Positives.

$$\text{Estimated Precision} < 20\% \tag{2}$$

The "Strict Global Locking" mechanism, while designed to prevent conflicts, ultimately locked onto incorrect neighbors early in the process. This suggests that the feature descriptors (normalized contour + average color) lacked the discriminative power to distinguish between the hundreds of visually similar "tab" shapes in the puzzle.

Fig. 6. Final Failed Corner Detections. Failures on some complex 3/4-tab pieces, where the model struggles to define the geometry.

*2) Recall:*

$$\text{Recall} \approx \frac{898}{1935} \approx 46.4\% \tag{3}$$

The system identified 898 unique connections. However, given the low precision observed above, this recall is inflated by false positives. The effective recall (valid connections found) is likely significantly lower, resulting in a graph that is not only sparse but structurally incorrect.

### D. Assembly Metrics

The final assembly quality was assessed based on the coherence of the reconstructed regions.

*1) Largest Connected Component (LCC):* The largest coherent cluster successfully assembled contained little success. Although the graph contained 898 edges, the connectivity was not uniform. The graph formed many small, disjointed groups (typically 2–4 pieces) rather than a large puzzle. This indicates that while the local shape matching was often succeeding, the feature descriptors lacked the discriminative power to bridge these small clusters into a larger whole.

*2) Visual Correctness and Drift:* Visual inspection of the final assembly reveals significant geometric drift. Chains of pieces contained small angular errors in the pairwise transformations. Without a global optimization framework to distribute this error, the alignment deviated significantly after just a few connections. The overall assembly algorithm was unable to merge these islands, resulting in a scattered reconstruction consisting of many fragmented blocks of pieces which were not necessarily correctly matched as seen in the final image.

## IV. Reflection

### A. Analysis of Successes

Despite the incomplete final reconstruction, several individual components of the pipeline demonstrated strengths.

- **Corner Detection Successes:** As seen in the corner identification, many pieces of a variety of pieces were

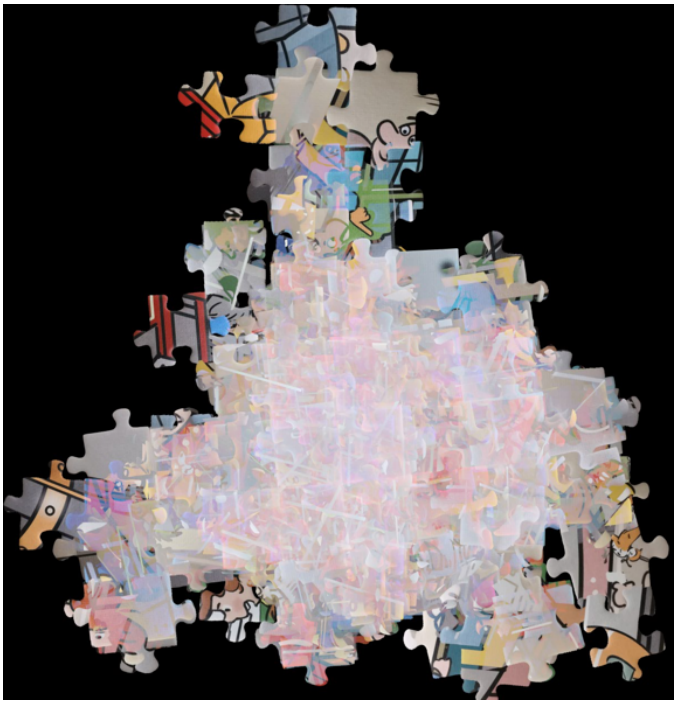

Fig. 7. Final Failed Reconstruction.This shows not only a failure in connectivity, but also challenges in matching corners to each other.

correctly processed. This confirms that the deep learning approach showed strong improvements. The system correctly solved the geometry for most pieces

- **Segmentation Strength:** The switch to EfficientNet-B4 with Test Time Augmentation provided masks that were sharp enough to allow for tight locking between pieces without significant pixel overlap or gaps.

### *B. Root Cause of Assembly Failure*

The failure to reconstruct the full 1000-piece image can be attributed to two primary factors:

- **Feature Ambiguity:** The 50-point normalised contour descriptor proved insufficient for a 1000-piece scale. In a set of this size, dozens of tabs share nearly identical geometric profiles. Without texture matching at the pixel level (eg. SIFT/ORB feature matching across the boundary), geometry alone generated too many false positives.
- **Greedy Locking Error:** The Strict Locking algorithm made the issue worse. Once an incorrect match was locked, it permanently removed both from the pool of available candidates, triggering a chain reaction of missing connections for the actual neighbors.

### *C. Future Work*

To bridge the gap between successes and failures, future iterations would implement:

1) **A Larger Training Set:** Using a larger set of images to train the model would be essential in improving the overall results. Feeding 5000+ labelled puzzle pieces into the model would likely improve accuracy across the board of the task.
2) **Global Relaxation:** Formulating the assembly as a global optimization problem (e.g., minimizing total energy in a Mass-Spring system) would distribute error across the entire grid, correcting drift.
3) **Soft Matching:** Replacing binary "Locking" with a probabilistic graph where edges have confidence weights, allowing the assembly algorithm to backtrack and break weak links if a better global configuration is found.

## V. Conclusion

This project presented an end-to-end computer vision pipeline designed to reconstruct a 1000-piece puzzle from unlabelled images. The investigation successfully demonstrated the power of deep learning in segmentation and feature extraction, with the EfficientNet-B4 U-Net achieving a validation IoU of 0.9787 and the heatmap regression model robustly identifying geometric keypoints for the majority of standard pieces. However, the transition from local features to global assembly proved to be the system's critical bottleneck. While the strict graph construction and pairwise affine transformation logic were mathematically sound, the lack of global error correction and the inherent brittleness of local shape matching resulted in a fragmented final reconstruction. The project highlights that while modern neural networks excel at semantic understanding (segmentation and corner detection), large-scale geometric reconstruction requires sophisticated frameworks to reduce errors and ensure structural coherence.

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Fig. 8. *
Figure 1: Final Full-Page Reconstruction of the 1000-Piece Mystery Puzzle.